EX NO. : 1

DATE:

Install Virtualbox / VMware Workstation with different flavours of linux or windows OS on top of windows 10 or 11.

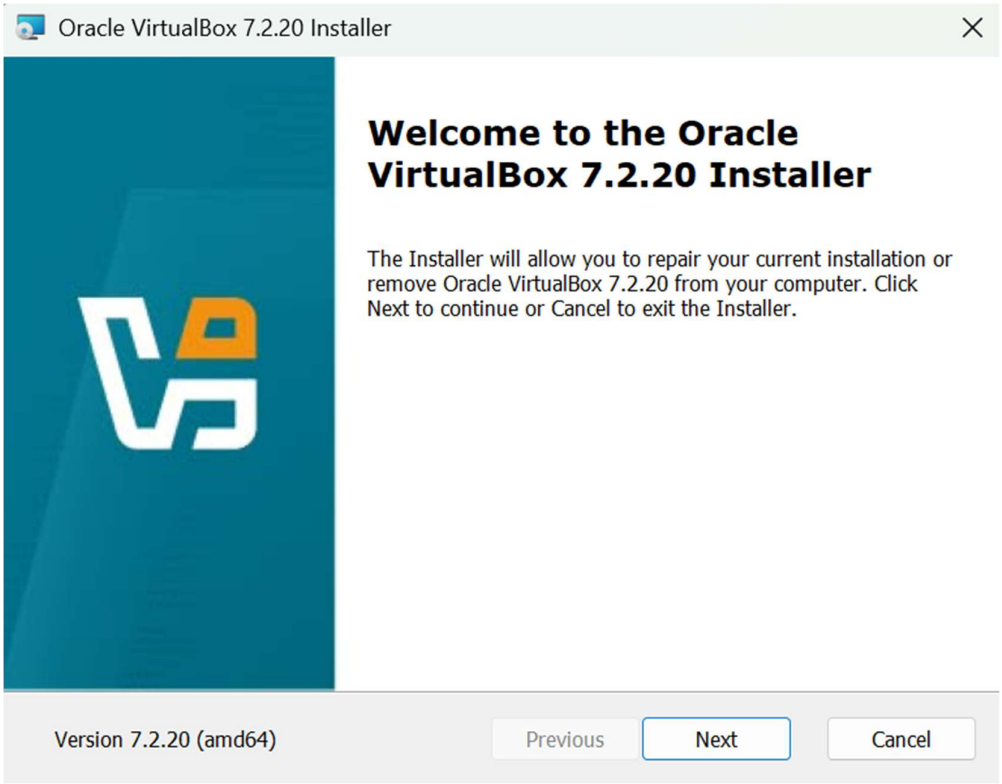
AIM:

To Install Virtualbox / VMware Workstation with different flavours of linux or windows OS on top of windows 10 or 11.

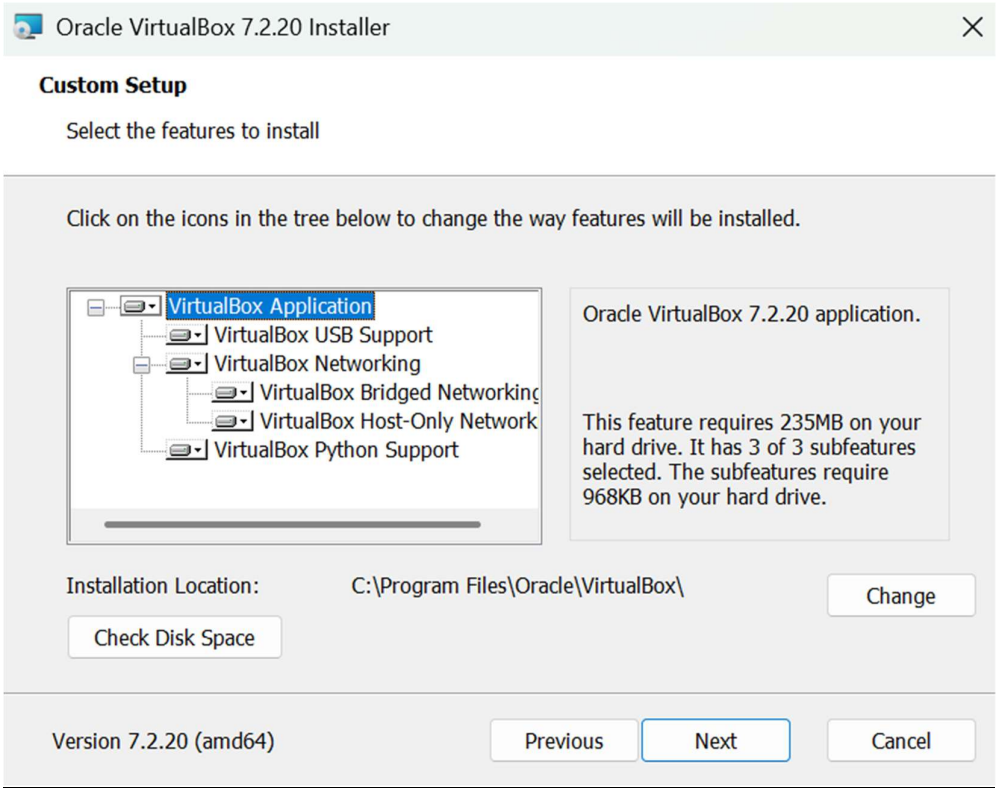
PROCEDURE:

Steps to install Virtual Box:

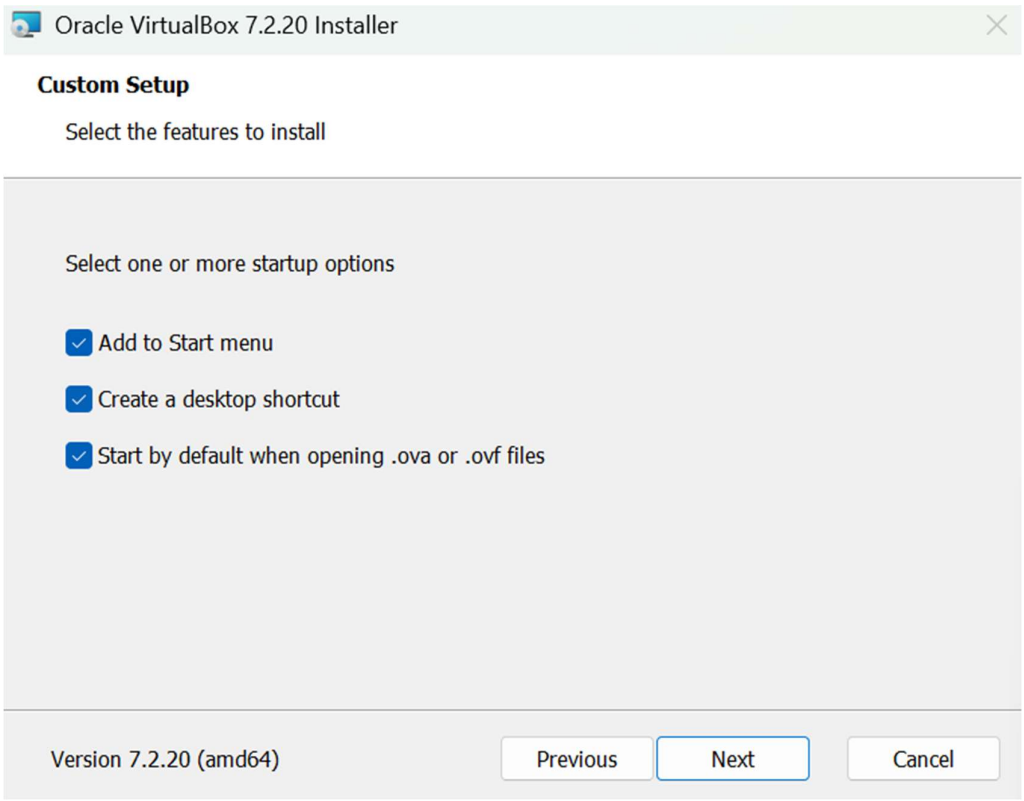
1. Download the Virtual box exe and click the exe file…and select next button.



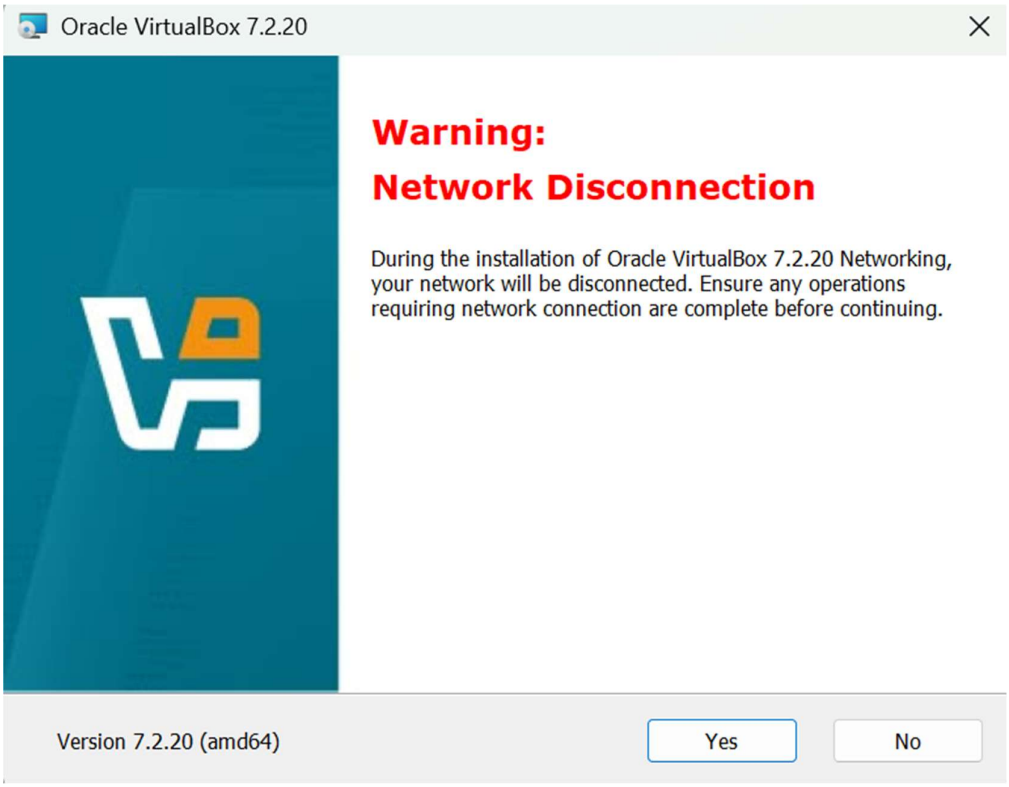
2. Click the next button.



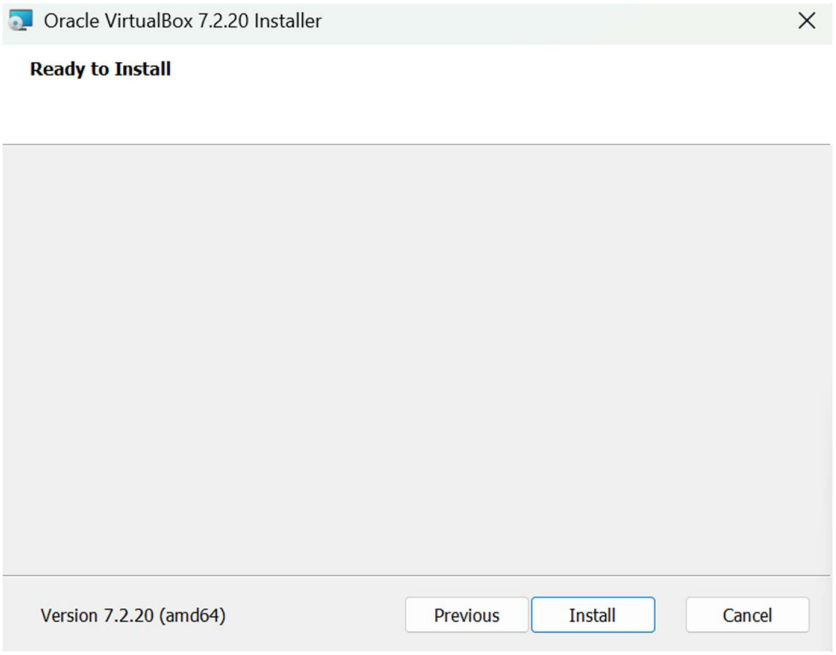
3. Click the next button.



4. Click the YES button.



5. Click the install button.

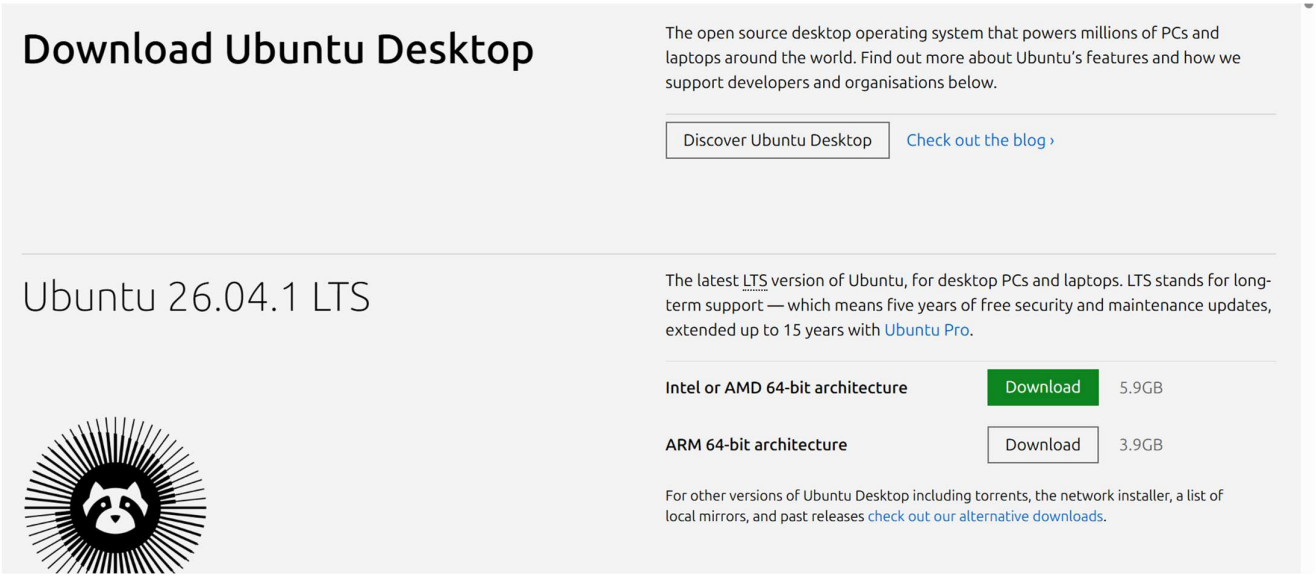


6. Then installation was completed,then it shows virtual box icon on desktop screen.

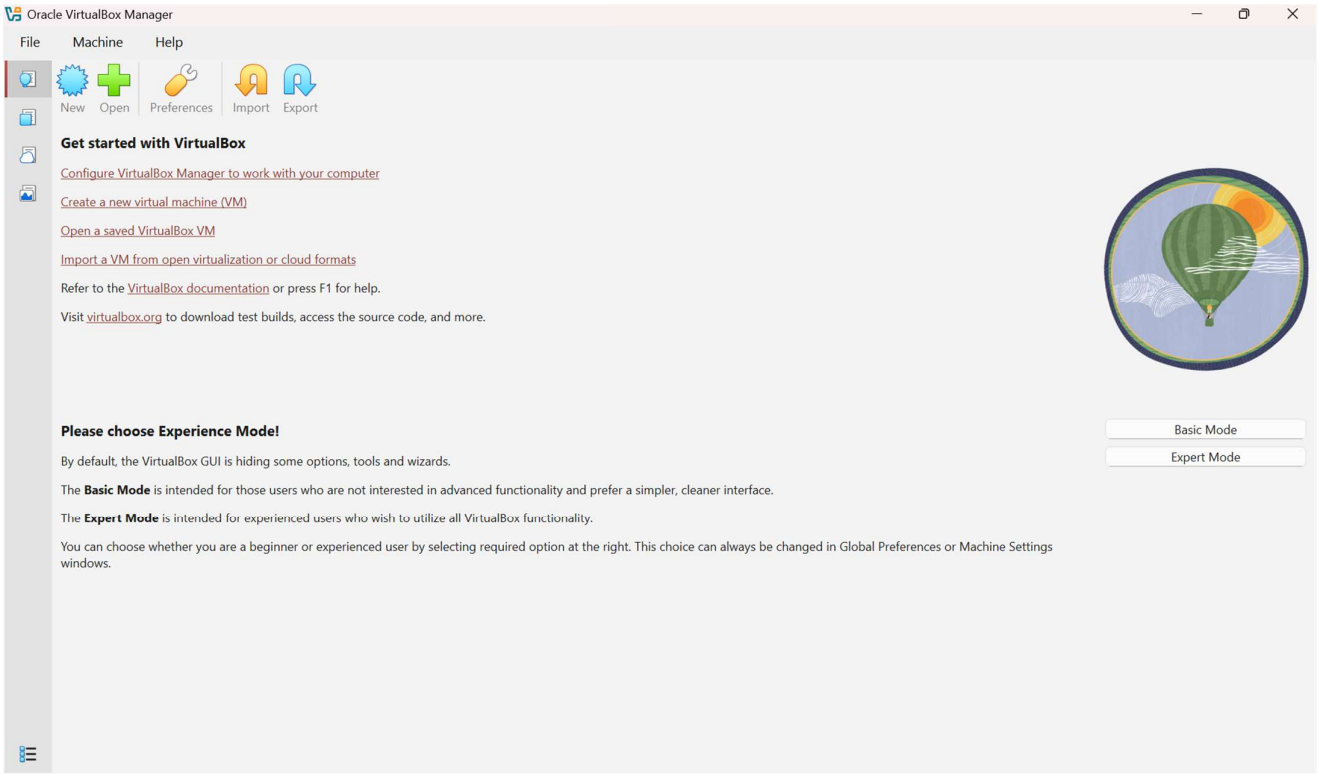


Steps to Open Opennebula:

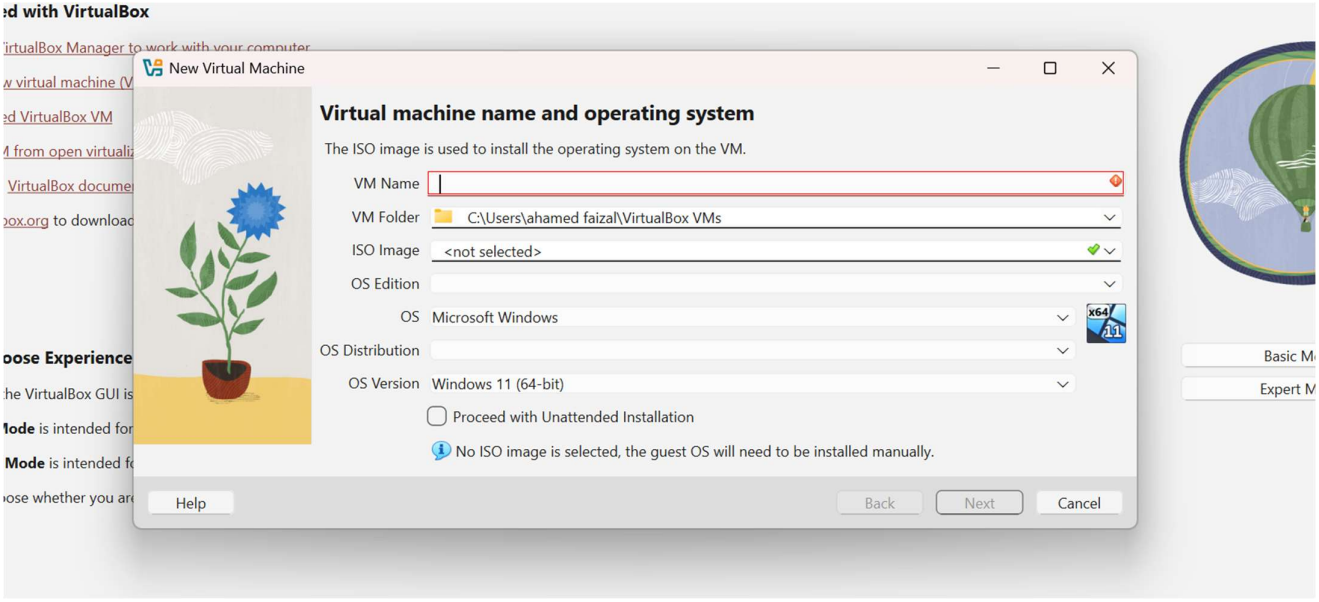
1. Install Ubuntu desktop



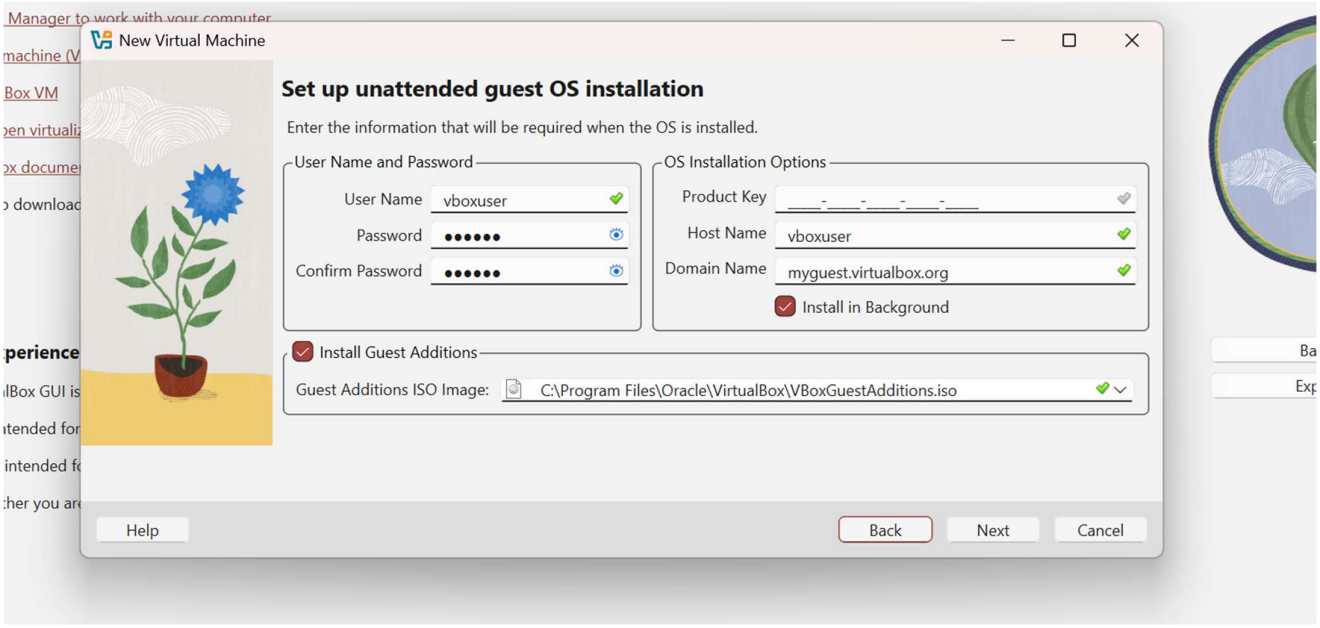
2. Open oracle virtual box Manager



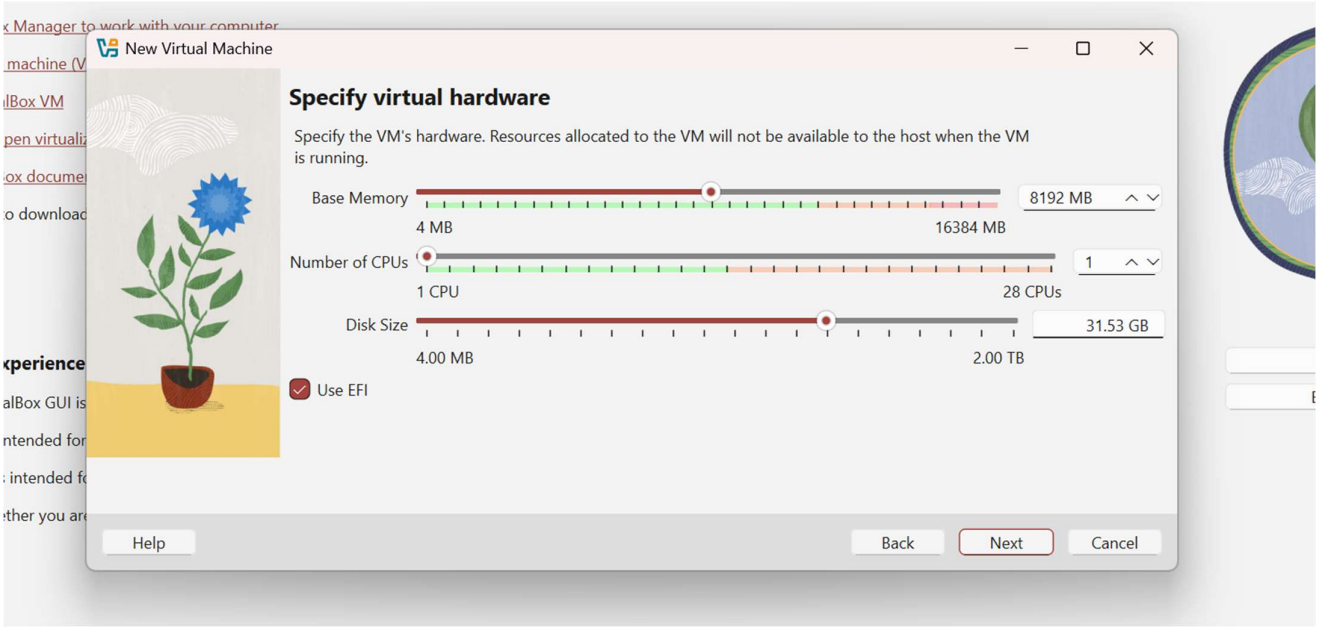
3. Click new -> type VM name , choose folder, in ISO image you choose the installed Ubuntu desktop



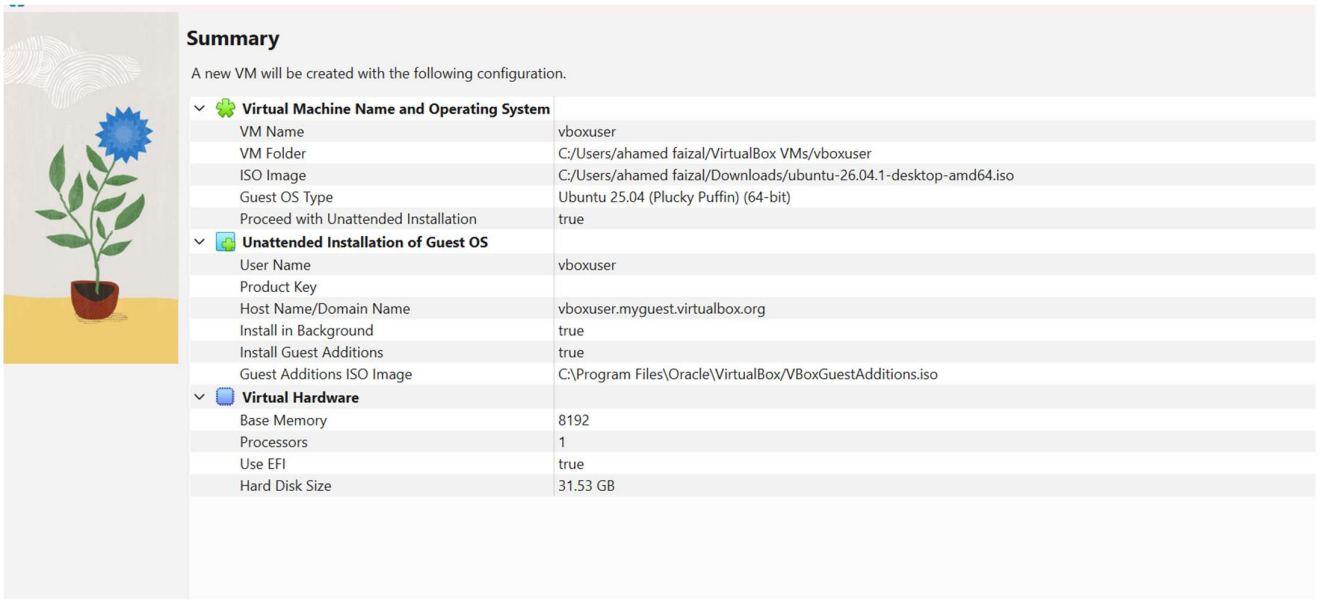
4. Click next then fill username password and host name select the two checkin boxes



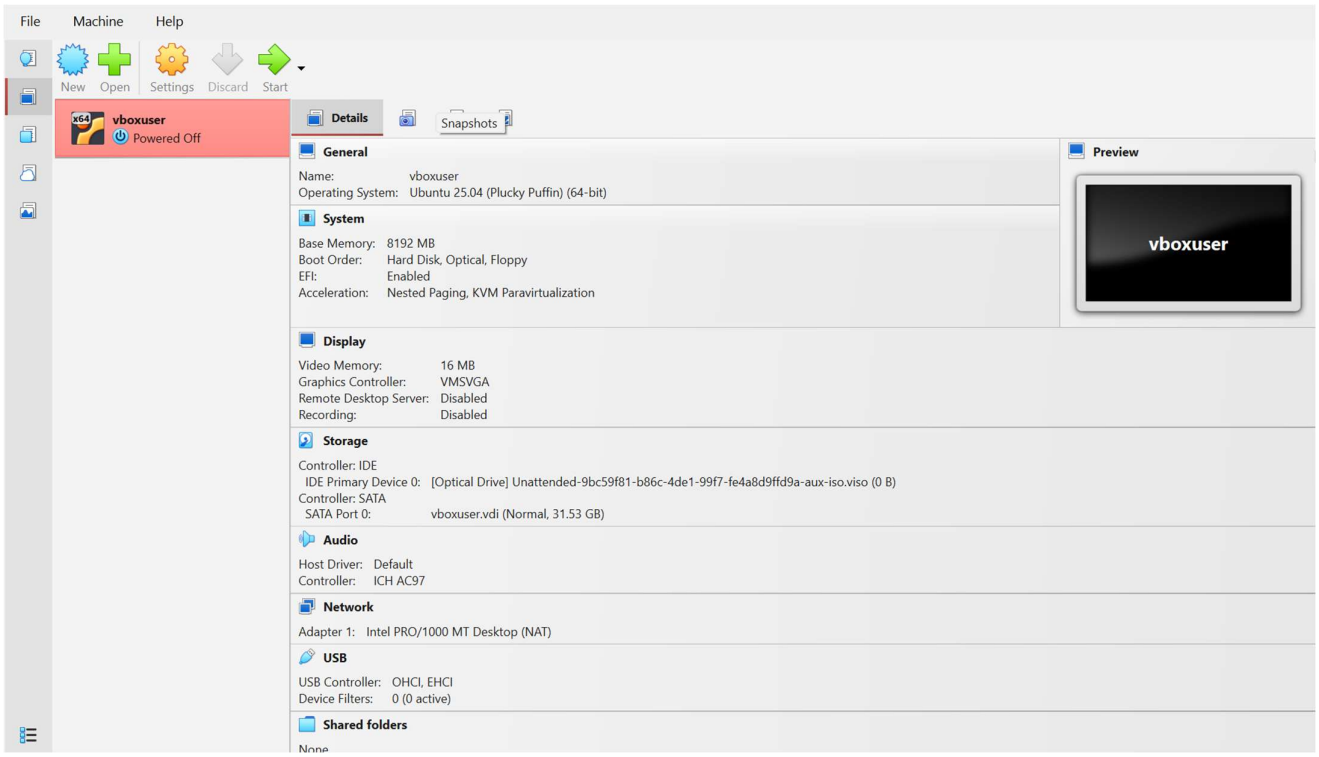
5. Click next select base memory nearly 8192mb , number of cpus select 4 , disk size should be 30gb or more than 30gb , select check in box



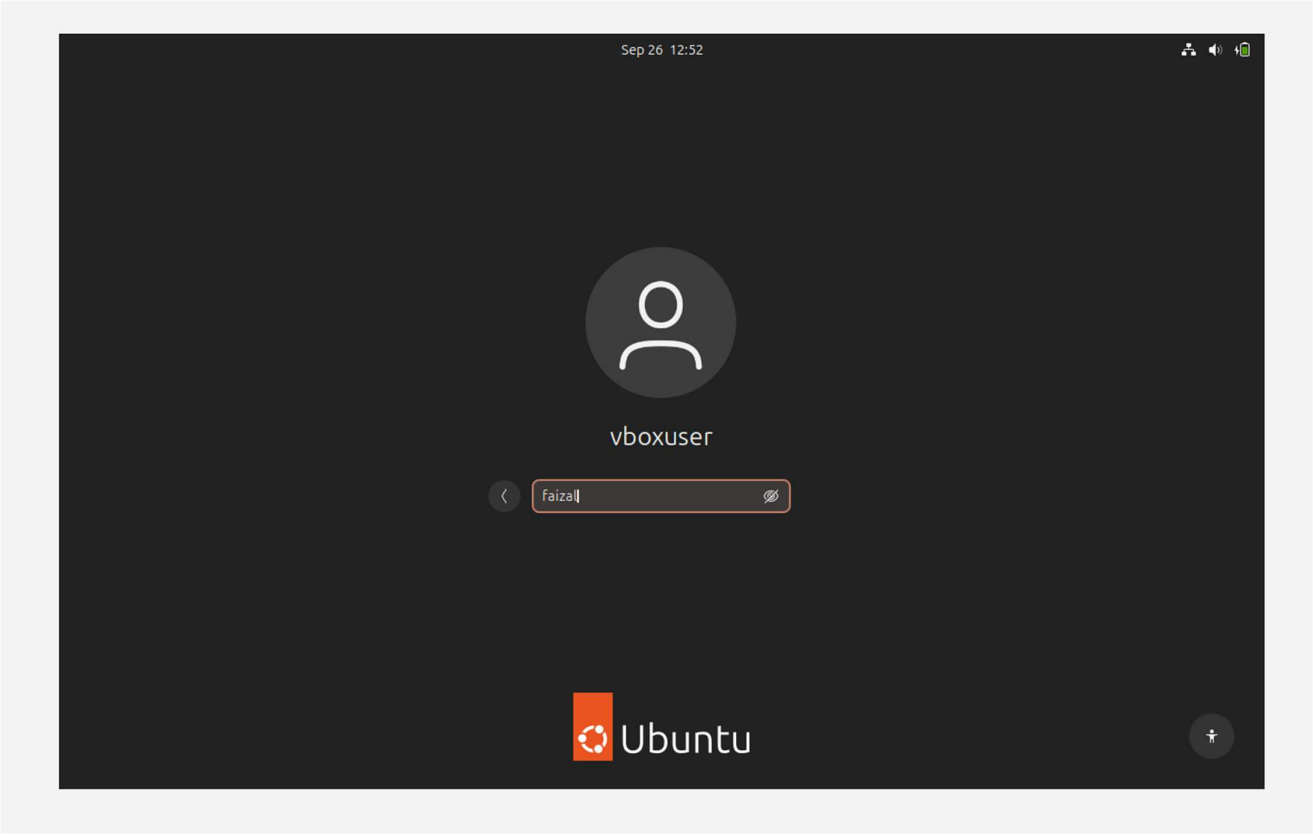
6. In summary your created vm details will be show check and give finish it will display the details of VM created and show in machine



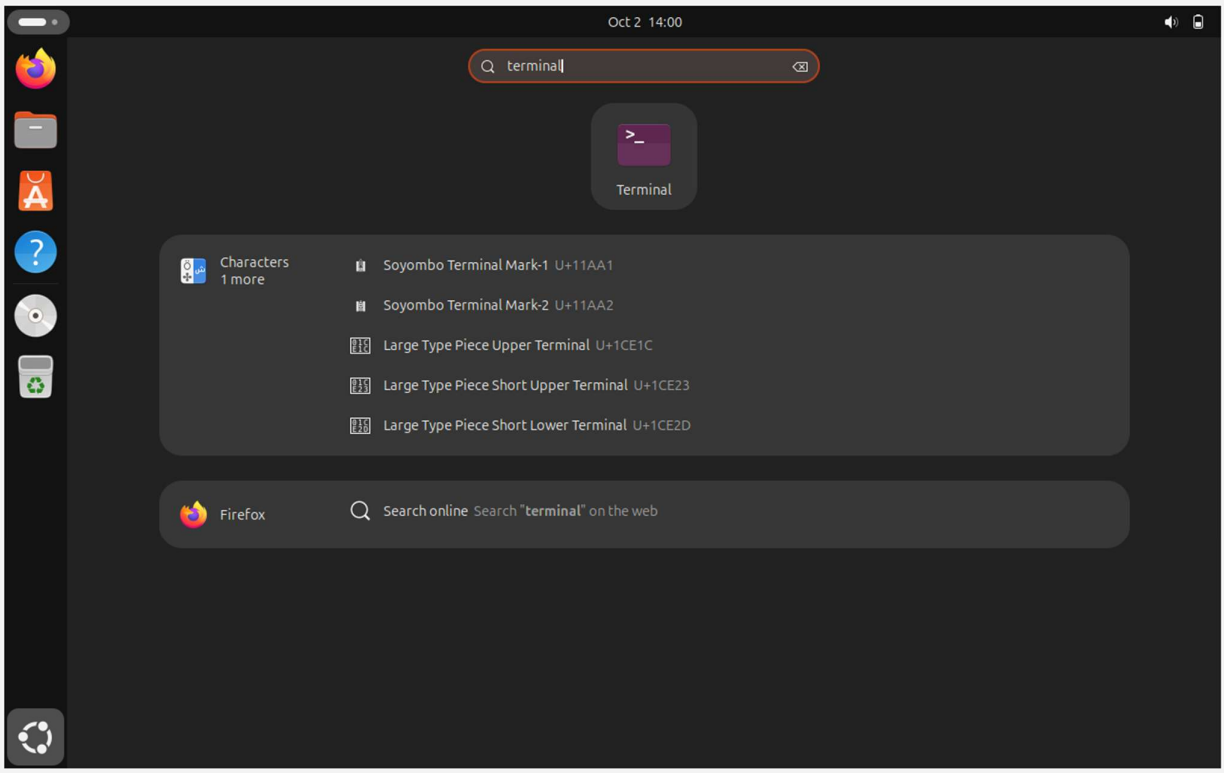
7. Start the vm it will run and open the Ubuntu desktop



8. After Ubuntu download , it will ask password.



9.In show apps open terminal



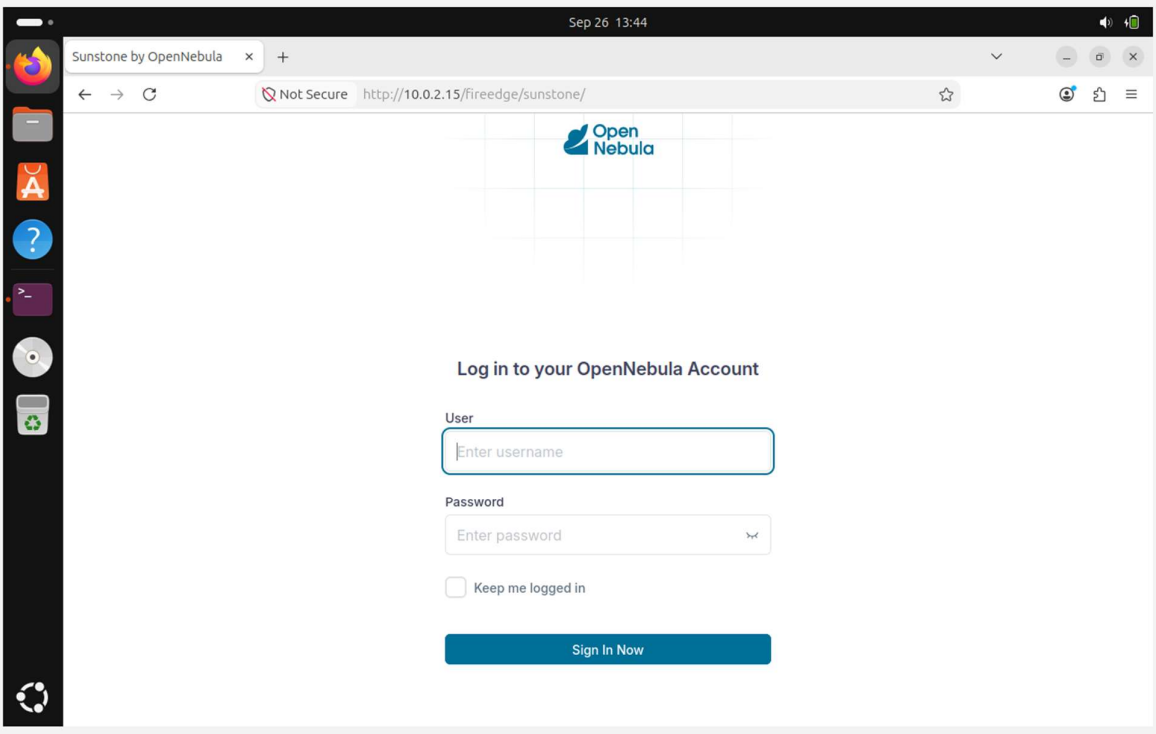
10. Type these command: For someone with the same Ubuntu/VirtualBox setup, the short version is:

- sudo apt update
- enter password : you already give some password in 2nd step enter that password
- sudo apt install -y openssh-server
- sudo systemctl enable –now ssh
- sudo apt install -y cpu-checker
- sudo apt install -y qemu-system-x86
- wget 'https://github.com/OpenNebula/minione/releases/download/v7.4.0/minione'
- chmod +x minione
- ip -br addr
- sudo ./minione \

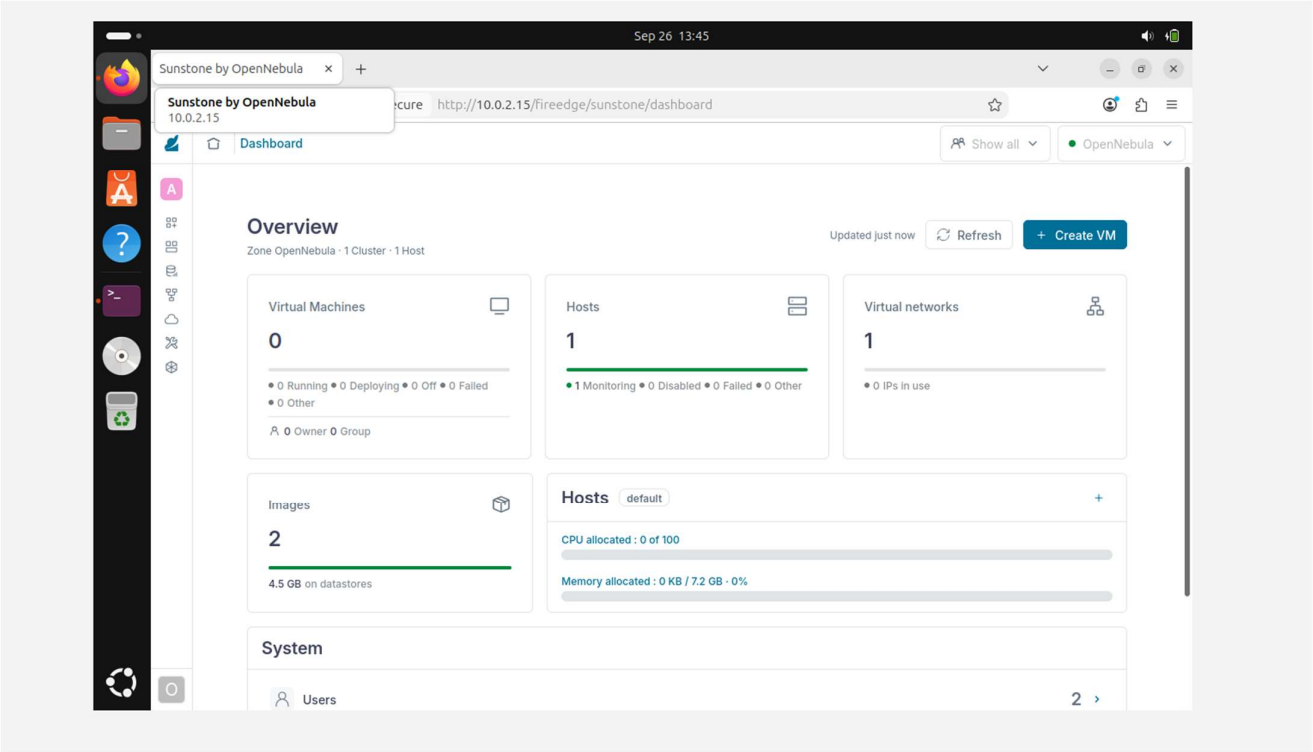
  >> --yes

- sudo cat /var/lib/one/.one/one_auth
- sudo systemctl start opennebula
- sudo systemctl status opennebula –no-pager

11. After type all the above commands, open firefox browser type in search box : http://10.0.2.15/



12. For username and password, this command : sudo cat /var/lib/one/.one/one_auth , shows the username and password . Enter that username and password in login page and sign in , it shows admin dashboard



APPLICATIONS:

There are various applications of cloud computing in today's network world. Many search engines and social websites are using the concept of cloud computing like www.amazon.com, hotmail.com, facebook.com,linkedln.com etc. the advantages of cloud computing in context to scalability is like reduced risk , low cost testing,ability to segment the customer base and auto-scaling based on application load.

RESULT:

Thus the procedure to run the virtual machine of different configuration.

EX.NO.:2

DATE:

Install a C compiler in the virtual machine created using virtual box and execute Simple Programs

AIM:

To Install a C compiler in the virtual machine created using virtual box and execute Simple Programs.

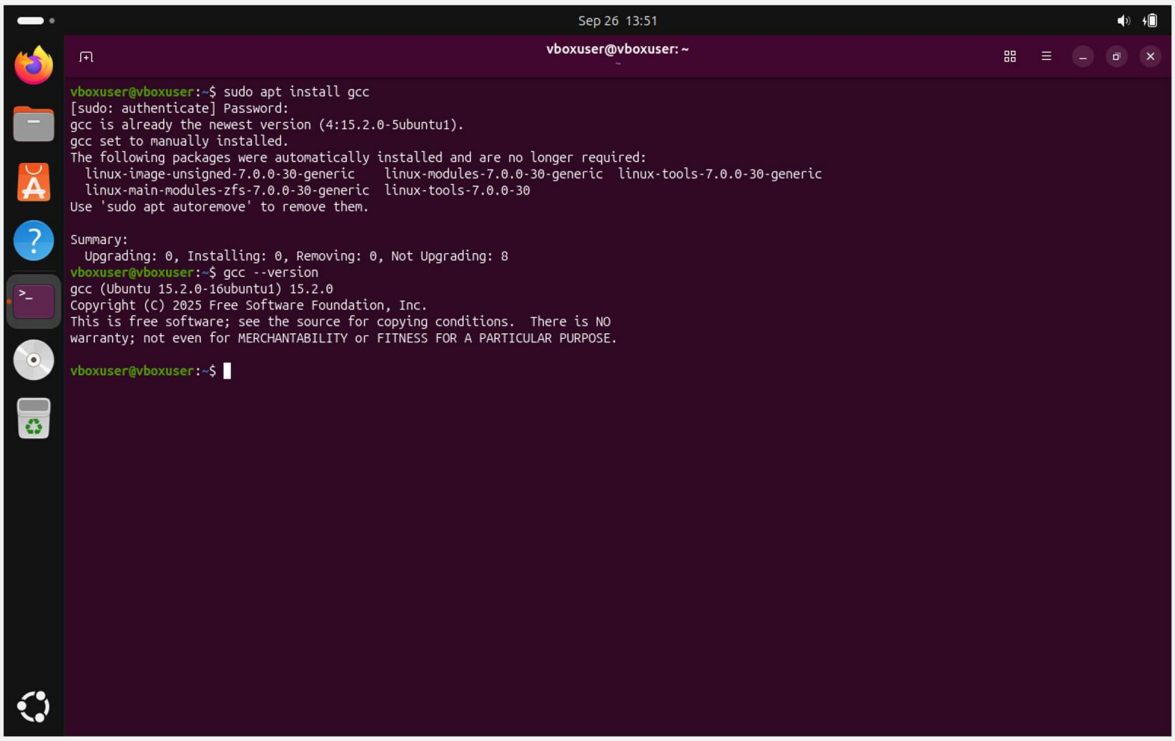
PROCEDURE:

1. Open terminal type this command: sudo apt update
2. Type: sudo apt install gcc
3. Check version: gcc --version
4. Type : nano hello.c
5. Type this program :

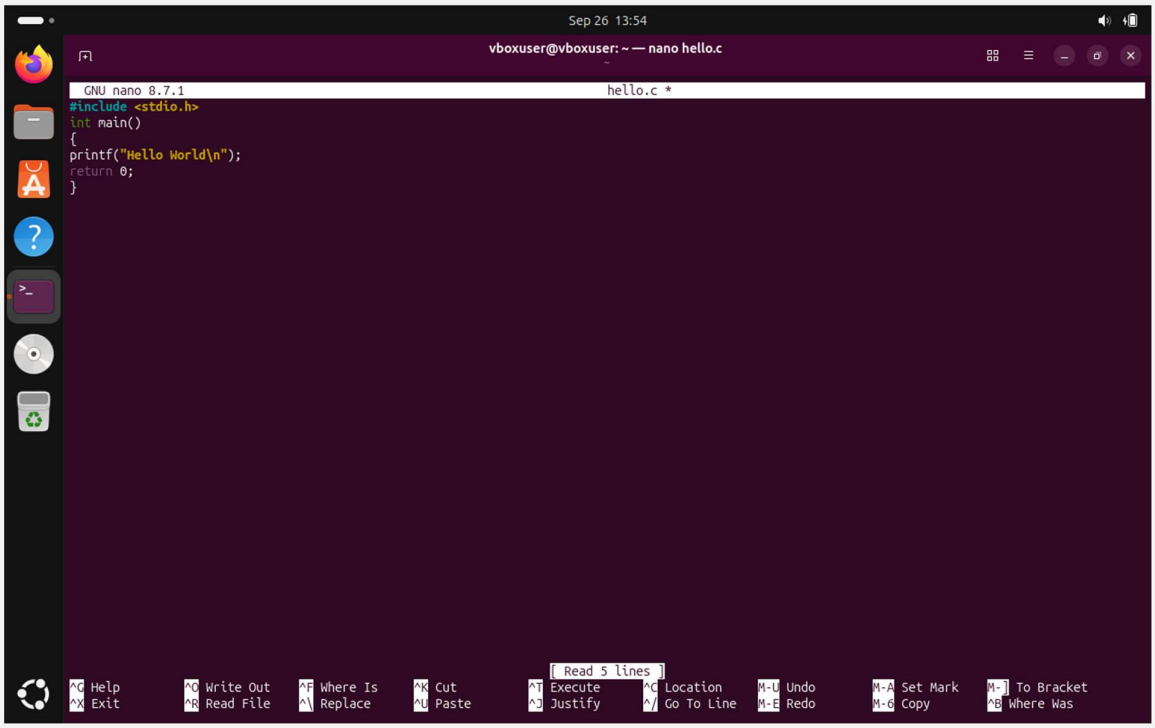
```
#include <stdio.h>
int main()
{
   printf("Hello World\n");
   return 0;
}
```

6. Ctrl + O → Enter → Ctrl + X
7. Compile : gcc hello.c
8. Run : ./a.out
9. Output: Hello World

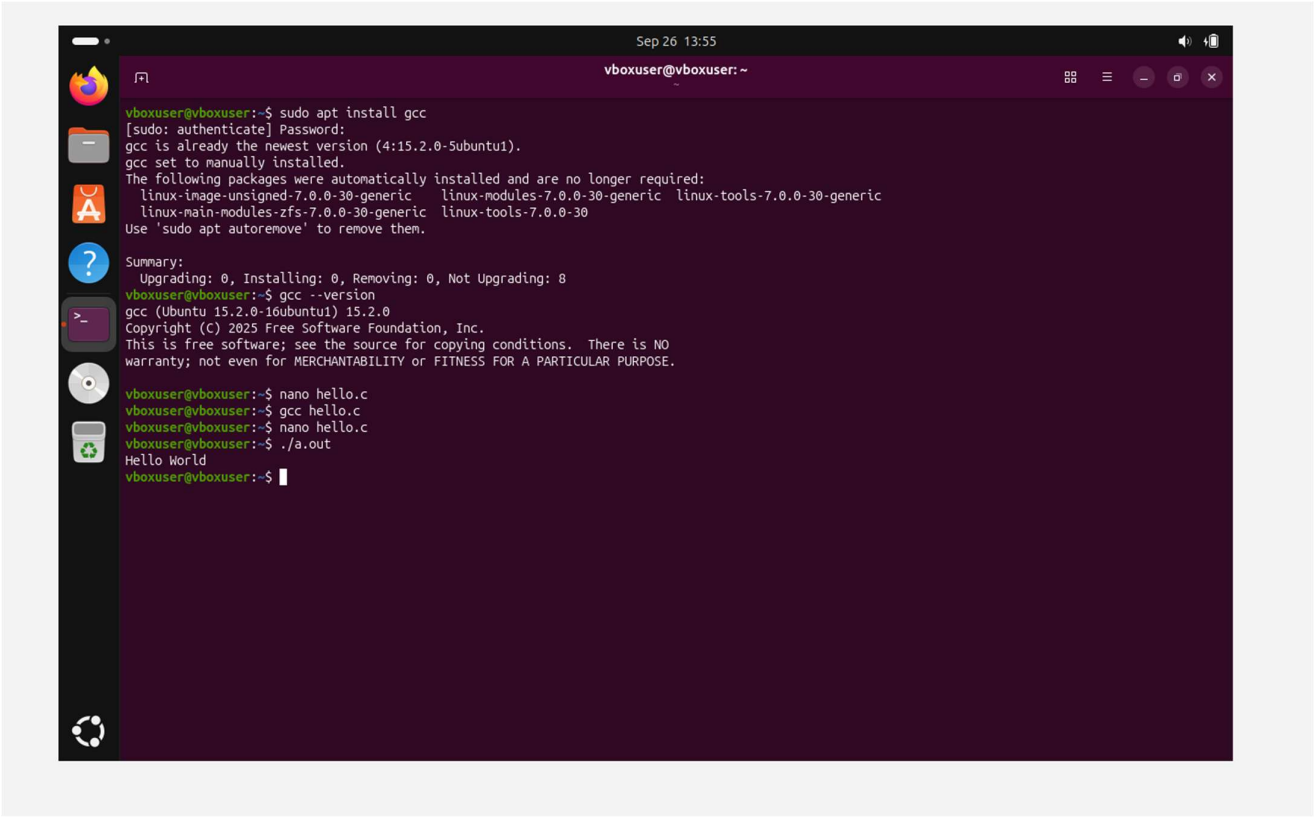
1. Type : nano hello.c



2. Type the C program :



3. Run the C program and get the output :



APPLICATIONS:

Simply running all programs in grid environment.

RESULT:

Thus the simple C programs executed successfully.